

Automated Customer Review Analysis: Final Report

Executive Summary

This report presents an end-to-end NLP system that converts raw Amazon product reviews into actionable marketing insights. The architecture comprises a sentiment classifier, an unsupervised product clustering model, and a generative review summarizer, all exposed via a full-stack web application (FastAPI backend, Next.js frontend).

1. Data Pipeline

The dataset merges three Kaggle/Datafiniti Amazon review exports into 59,743 unique reviews spanning 89 raw product IDs across 7 Amazon-centric hardware brands (Kindles, Fire tablets, Echo devices).

- **Data Quality Fix:** Product IDs were unreliable for ~11% of reviews due to string concatenation and identifier collisions (e.g., a single ID spanning tablets, chargers, and grocery items). A cleaned product name was used as the primary entity identifier instead.
- **Class Distribution:** Highly skewed toward positive sentiment (~69% 5-star reviews).

2. Sentiment Classification

Ratings were mapped to 3 classes (1–2 stars: negative, 3: neutral, 4–5: positive) and evaluated zero-shot/pretrained on a held-out test set (n=500).

Model	Accuracy	Macro F1	F1 (Neg)	F1 (Neu)	F1 (Pos)
OpenAI (gpt-4o-mini)	91.6%	56.4%	0.622	0.108	0.963
OpenAI (gpt-5.4-mini)	90.6%	55.5%	0.609	0.098	0.957
Local RoBERTa	90.0%	52.8%	0.541	0.091	0.953
Random	94.4%	40.7%	0.250	0.000	0.971

Model	Accuracy	Macro F1	F1 (Neg)	F1 (Neu)	F1 (Pos)
Forest (TF-IDF)					

Key Findings: Macro F1 is the key benchmark due to class imbalance. OpenAI's gpt-4o-mini led overall (56.4% Macro F1). All models struggled on neutral (3-star) reviews due to sample sparsity and conflicting sentiment in text. Prompt engineering attempts did not improve neutral recall.

3. Product Clustering

Product names were embedded using all-MiniLM-L6-v2 and clustered into 5 distinct categories via K-Means:

Category	Scope
Fire Tablets	Fire HD, Kindle Fire, and Kids Edition tablets.
E-Readers & Accessories	Kindle e-readers (Paperwhite, Oasis) and cases.
Echo & Smart Speakers	Echo, Echo Show, Echo Spot devices.
Fire TV & Accessories	Fire TV Sticks, media players, remotes.
Speakers & Power Accessories	Amazon Tap, power adapters, and chargers.

4. Review Summarization

- **Architecture:** Document structure and Bayesian product rankings (m=50) are generated programmatically to prevent hallucination. The LLM/BART model is restricted strictly to condensing complaint snippets into single-sentence summaries.
- **Backends:** Configured to use BART locally for reproducible testing and OpenAI (gpt-4o-mini) in production.

5. Deployment & Next Steps

- **Infrastructure:** FastAPI on Render (backend) and Next.js on Vercel (frontend). To fit within Render's free 512MB RAM limit, production runs API calls instead of loading PyTorch

weights.

- **Future Improvements:** Dedicated neutral-class classifier, rule-based overrides for clustering boundary cases, and expanding summarization to cover positive product highlights.